Physics GCE A Level 2023 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0780 Physics 1, June 2023, Advanced Level, one and a half hours). Answer ALL 50 questions; all questions carry equal marks. Each question has FOUR suggested answers A, B, C and D; mark only one answer for each question (if you mark more than one you score zero). Mobile phones are NOT allowed in the examination room. The use of a calculator and the Formulae book is allowed.

Constants used in the answers (the paper itself prints no data sheet): acceleration of free fall $g = 9.81\ \mathrm{m\,s^{-2}}$ (question 14 uses the value $10\ \mathrm{N\,kg^{-1}}$ given in its text); specific heat capacity of water $4200\ \mathrm{J\,kg^{-1}\,K^{-1}}$; specific latent heat of vaporisation of water $2.26 \times 10^{6}\ \mathrm{J\,kg^{-1}}$; speed of light $c = 3.00 \times 10^{8}\ \mathrm{m\,s^{-1}}$; Planck constant $h = 6.63 \times 10^{-34}\ \mathrm{J\,s}$.

Section I (thirty five questions), Questions 1-35. Directions: Each of the thirty five questions or incomplete statements in this section is followed by four suggested answers. Select the best answer in each case.

Section II (ten questions), Multiple Selection, Questions 36-45. Directions: for each group of questions below ONE or TWO of the responses given is/are correct. Then choose: A - if 1 and 2 are correct; B - if 2 and 3 are correct; C - if 1 only is correct; D - if 3 only is correct. (Directions summarised: A = 1, 2 only; B = 2, 3 only; C = 1 only; D = 3 only.)

Section III (five questions), Questions 46-50. Directions: Each of the questions (46-50) has four sets of graphs A-D. Which of the graphs in each question best fits the relationship between x and y.

1. Which of the following consists of only scalar quantities?

a) Electric field intensity, mass, and volume
b) Momentum, acceleration and displacement
c) Charge, work, temperature
d) Speed, electromagnetic force, distance

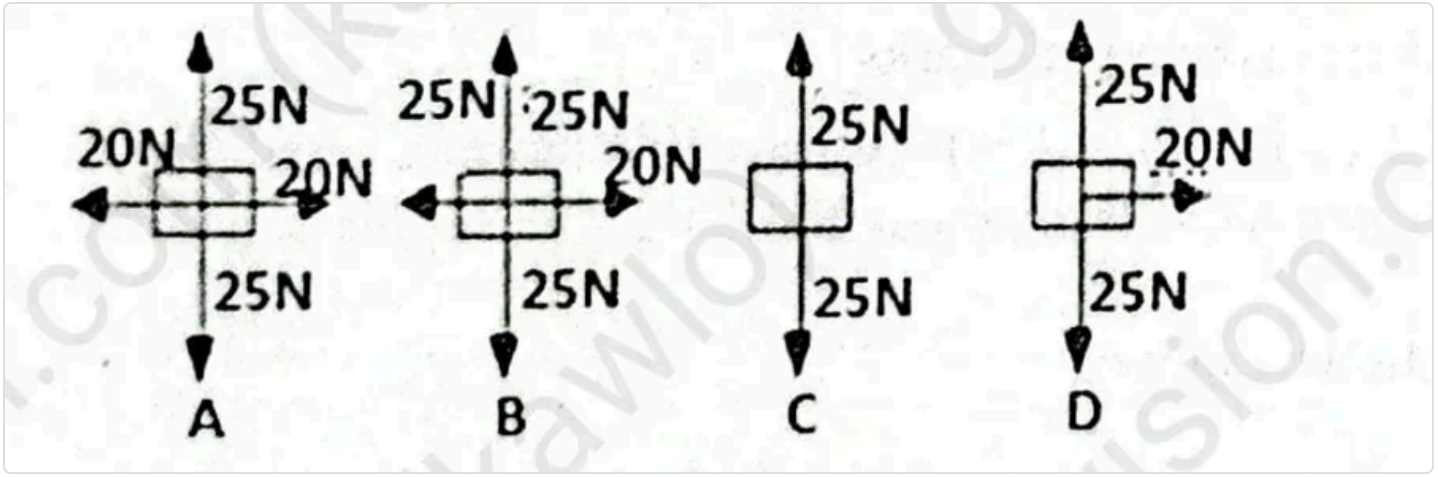
2. Rutherford's alpha-particle scattering experiments reveal strongly that an atom

a) Has a nucleus of very small size and very high density.
b) has an evenly distributed mass
c) has a nucleus of relatively large size and of low density
d) is a positive mass with electrons spreading all over inside to make it neutral

3. A ball bearing was made to fall freely through a viscous liquid in a very long tube. State which of the following statements is true?

Options as printed: A) The acceleration of the ball remains constant throughout. B) The net upward force remains constant. C) The net downward force remains constant. D) The velocity of the ball remains constant throughout

4. Which of the force diagram(s) show(s) an object moving to the right with a constant velocity?



Options as printed: the diagrams A to D of the figure (A: 25 N up, 25 N down, 20 N left, 20 N right; B: the same as A with a duplicated 25 N label; C: 25 N up and 25 N down only; D: 25 N up, 25 N down and 20 N to the right).

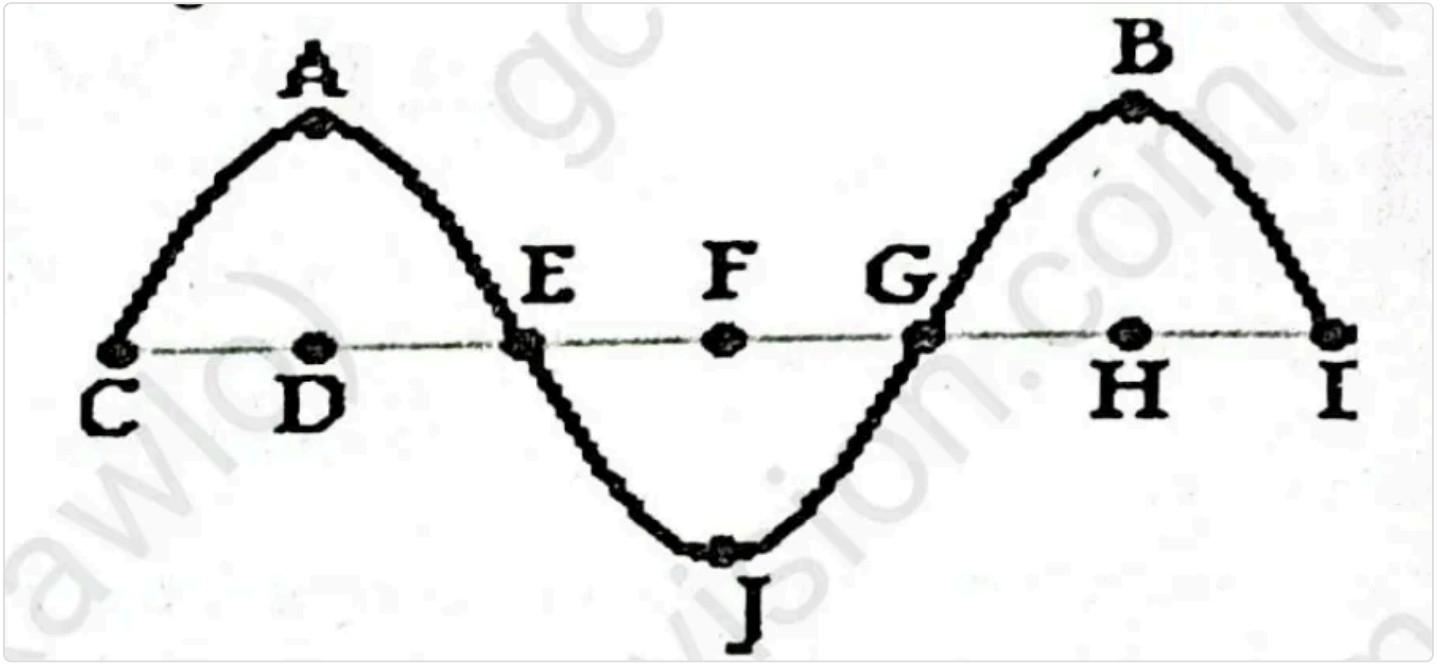
5. According to Newton's third law, every force is accompanied by an equal and opposite reaction force. The reason why these forces do not cancel each other is?

a) the action force acts for a longer time period
b) the two forces are not always in the same direction
c) one of the two forces is greater than the other
d) the two forces act upon different objects; only forces on the same object can balance each other.

6. A student applies a force of $26\ \mathrm{N}$, parallel to a plane surface inclined at 30° to the horizontal in order to pull an object of weight $40\ \mathrm{N}$ along the plane surface at constant speed. The frictional force between the plane and the object is

a) $6\ \mathrm{N}$
b) $24\ \mathrm{N}$
c) $26\ \mathrm{N}$
d) $30\ \mathrm{N}$

7. Figure 2 shows a transverse wave travelling through a medium to the right. See diagram below. The particles of the medium are vibrating.



a) perpendicular to the line joining AD.
b) parallel to the line joining AD.
c) along the line joining CI.
d) at various angles to the line CI.

8. Which of the following statements is true concerning a ball thrown vertically upwards and caught on its return by the thrower?

a) the ball takes a shorter time to go up than to come down.
b) the speed is constant throughout the flight.
c) the velocity is constant throughout the flight
d) the acceleration is constant throughout the flight.

9. A power cable carrying a current of $300\ \mathrm{A}$ is placed normally in a uniform B-field of flux density $10^{-5}\ \mathrm{T}$. If the cable experiences a force of $1.2\ \mathrm{N}$, then the length of the cable is

a) $40\ \mathrm{m}$
b) $400\ \mathrm{m}$
c) $4000\ \mathrm{m}$
d) $1200\ \mathrm{m}$

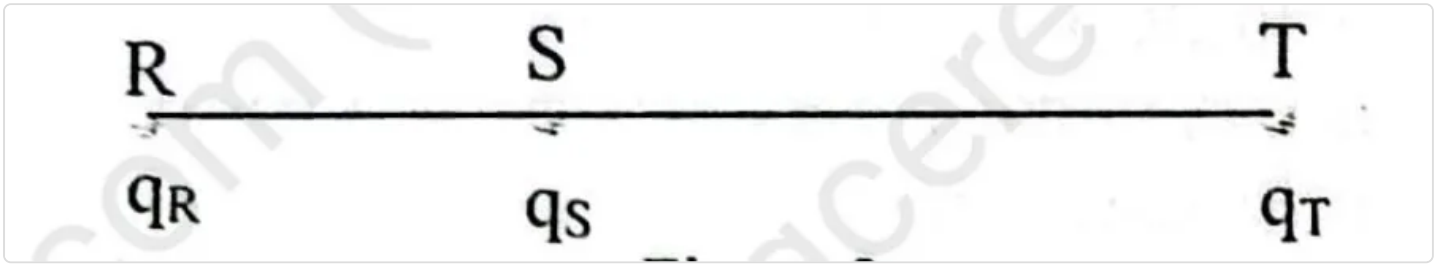
10. Which thermometer is best for measuring rapidly changing temperatures?

a) Liquid-in-glass
b) thermocouple
c) constant volume gas thermometer.
d) platinum resistance

11. The amount of energy needed to completely convert $1\ \mathrm{kg}$ of pure water at $60\ ^{\circ}\mathrm{C}$ to steam at $100\ ^{\circ}\mathrm{C}$ and at atmospheric pressure is

a) $1.68 \times 10^5\ \mathrm{J}$
b) $2.26 \times 10^6\ \mathrm{J}$
c) $2.09 \times 10^6\ \mathrm{J}$
d) $2.43 \times 10^6\ \mathrm{J}$

12. Figure 3 shows three charges of the same sign and magnitude placed at points R, S and T. The point at which the resultant electric field strength is zero is:



Options as printed: A) Between R and S B) Midway between S and T C) Between S and T but closer to T D) Lies along the line joining R and T but behind T or B

13. A light ray which is incident on an air - water boundary at an angle of incidence different from zero. Which of the following statements is correct?

a) The ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant.
b) The wavelength of the light increases as it moves from air to water.
c) The speed of light increases as it moves from air to water.
d) The frequency of light increases as it moves from air to water.

14. The force on a $1\ \text{kg}$ mass on the surface of the earth is $10\ \text{N}$. If the radius of the earth is $6400\ \text{km}$, the force on $91\ \text{kg}$ mass at a point $4000\ \text{km}$ from the centre of the earth (assume the earth has uniform density) is

Options as printed: A) $1.6\ \text{N}$ B) $67.6\ \text{N}$ C) $16.0\ \text{N}$ D) $6.3\ \text{N}$

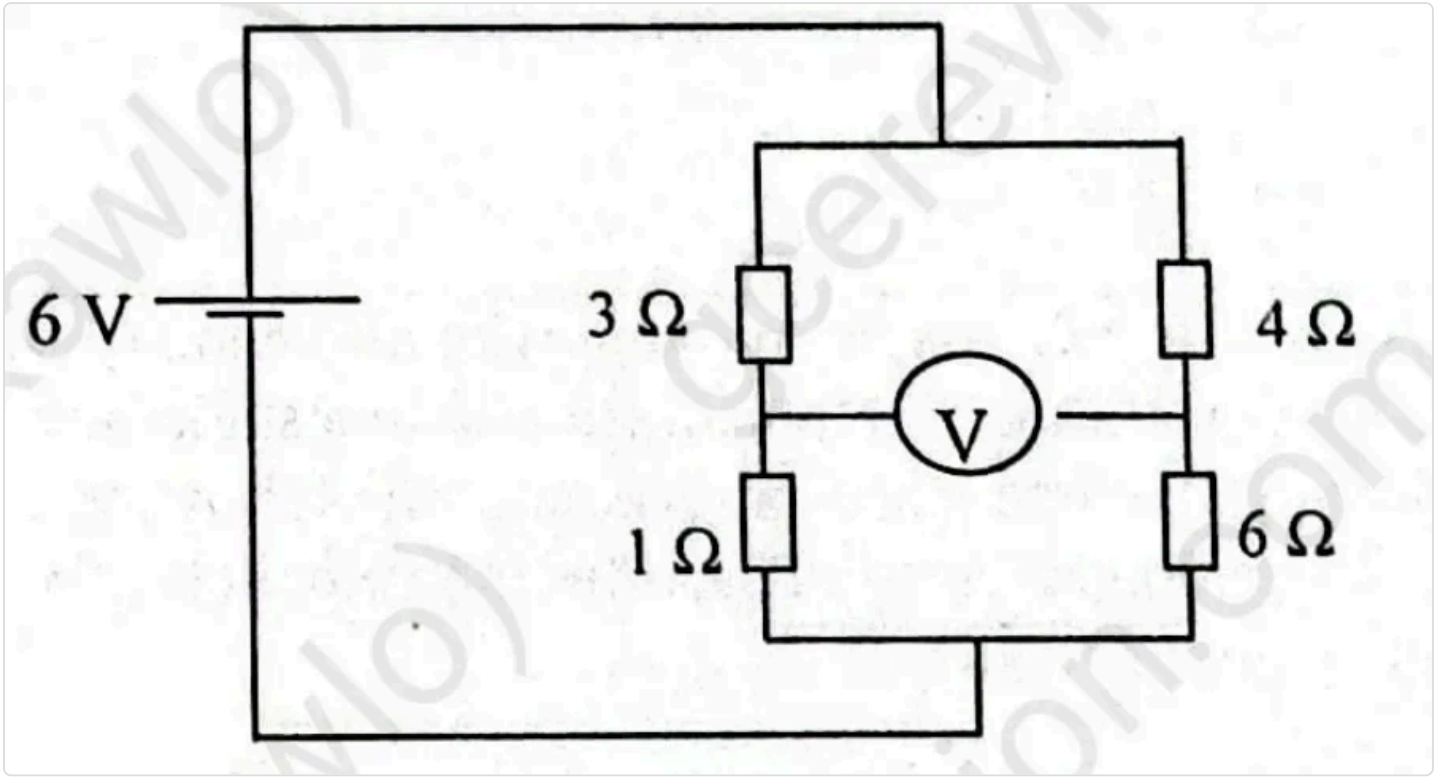
15. An a.c signal of $400\ \text{Hz}$ is applied to the y-plate of a CRO whose time-base is set at $2\ \text{ms cm}^{-1}$. The number of cycles on the $60\ \text{cm}$ wide screen of the CRO is

a) 12
b) 120
c) 48
d) 60

16. A train of mass m is travelling with a speed v. If the mass of the train is tripled by loading it, and if the velocity is halved, then how will the kinetic energy of the train vary?

a) Decrease by a factor of 3/4
b) Increase by a factor of 3/2
c) Remains the same
d) Decrease by a factor of 1/3

17. Figure 4 shows an electric circuit. The reading on V is



a) $1.5\ \text{V}$
b) $2.1\ \text{V}$
c) $3.6\ \text{V}$
d) $5.1\ \text{V}$

18. A vehicle is climbing a hill at constant velocity. Which of the following statements about the vehicle is true?

a) The rate of change of the vehicle's momentum is zero.
b) The net force acting on the vehicle is pointing upwards.
c) The frictional force between the vehicle and the surface is zero.
d) The force produced by the vehicle's engine is equal to the frictional force between the tyres and the road.

19. A stretched string is made to vibrate such that its third harmonic frequency is $2000\ \text{Hz}$. Its fundamental frequency is

a) $2000\ \text{Hz}$
b) $1000\ \text{Hz}$
c) $667\ \text{Hz}$
d) $500\ \text{Hz}$

20. A pure inductor has an inductance of $5 \times 10^{-2}\ \text{H}$. Calculate the rate of change of current through it that would produce an induced emf of $100\ \text{V}$.

a) $500\ \text{A s}^{-1}$
b) $2000\ \text{A s}^{-1}$
c) $200\ \text{A s}^{-1}$
d) $5\ \text{A s}^{-1}$

21. The pressure P, volume V and temperature of a real gas are related by the equation $\left(P + \frac{a}{V^2}\right)(V - b) = RT$, the unit of the constant, *"a"* is

a) $\text{kg m}^{-1}\ \text{s}^{-2}$
b) $\text{kg m}^{5}\ \text{s}^{-2}$
c) $\text{kg m}^{-2}\ \text{s}^{-2}$
d) $\text{kg m}^{-5}\ \text{s}^{-2}$

22. A body, x, of mass, m, moving with a velocity, v, makes an elastic head on collision with a stationary body, y, of same mass, m. Which of the following combinations correctly describes the velocities of x and y after collision.

a) x: v, y: 0
b) x: 0, y: v
c) x: $v/2$, y: $v/2$
d) x: 0, y: $v/2$

23. The half life of a radioactive isotope, X, is 10 hours. If the original activity is $20\ \text{Bq}$, its activity in Bq, after 30 hours will be

a) 10
b) 5
c) 2.5
d) 1.25

24. A battery of emf $10\ \text{V}$ and internal resistance 3 ohms is connected to a resistor. If the current in the circuit is $0.5\ \text{A}$, what is the resistance of the resistor and the p.d. across it when the circuit is closed?

a) 17 ohms and 8.5 V
b) 20 ohms and 10V

c) 8.5 ohms and 9V
d) 8.5 ohms and 9V

25. When a forward bias p.d is applied to a p-n junction, it;

a) lowers the barrier potential
b) raises the barrier potential
c) reduces the majority carrier current
d) increase number of carriers crossing the junction

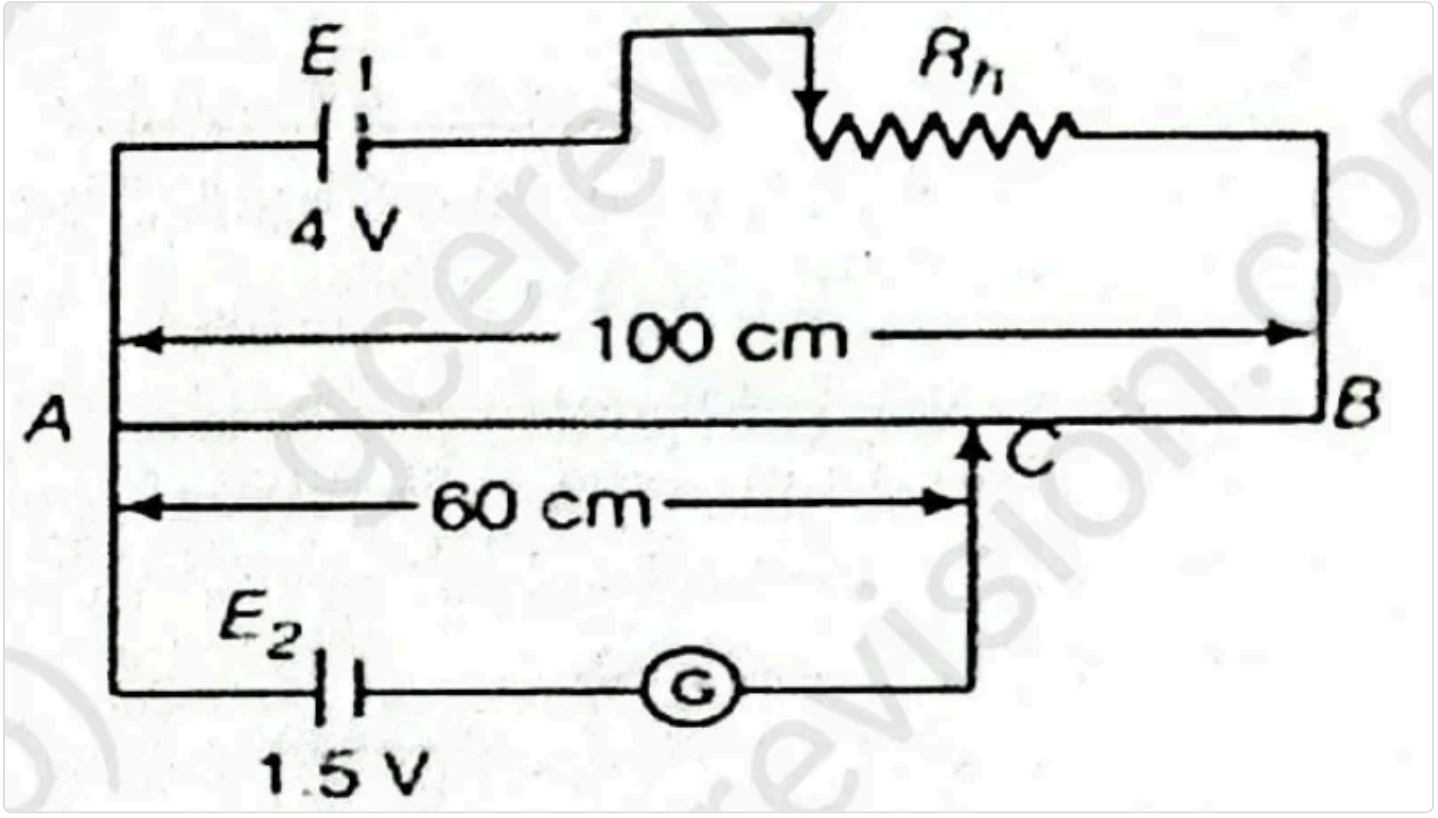
26. A silver wire has a resistance of 2.1 ohms at $27.5\ ^{\circ}\mathrm{C}$ and a resistance of 2.7 ohms at $100\ ^{\circ}\mathrm{C}$. The temperature coefficient of the silver wire is?

Options as printed: A) $0.02\ ^{\circ}\mathrm{C}^{-1}$ B) $0.05\ ^{\circ}\mathrm{C}$ C) $0.005\ ^{\circ}\mathrm{C}$ D) $0.009\ ^{\circ}\mathrm{C}^{-1}$

27. A $10\ \mathrm{kW}$ drilling machine is used to drill a hole in an aluminium block of mass $8.0\ \mathrm{kg}$. How much is the rise in temperature of the block in 2.5 minutes, given that the specific heat capacity of aluminum is $910\ \mathrm{J\,kg^{-1}\,K^{-1}}$

a) $103\ ^{\circ}\mathrm{C}$
b) $288\ ^{\circ}\mathrm{C}$
c) $1.7\ ^{\circ}\mathrm{C}$
d) $206\ ^{\circ}\mathrm{C}$

28. Figure 5 shows a potentiometer the test cell of emf $1.5\ \mathrm{V}$ is balanced at the point C. What is the p.d between the ends A and B of the potentiometer?

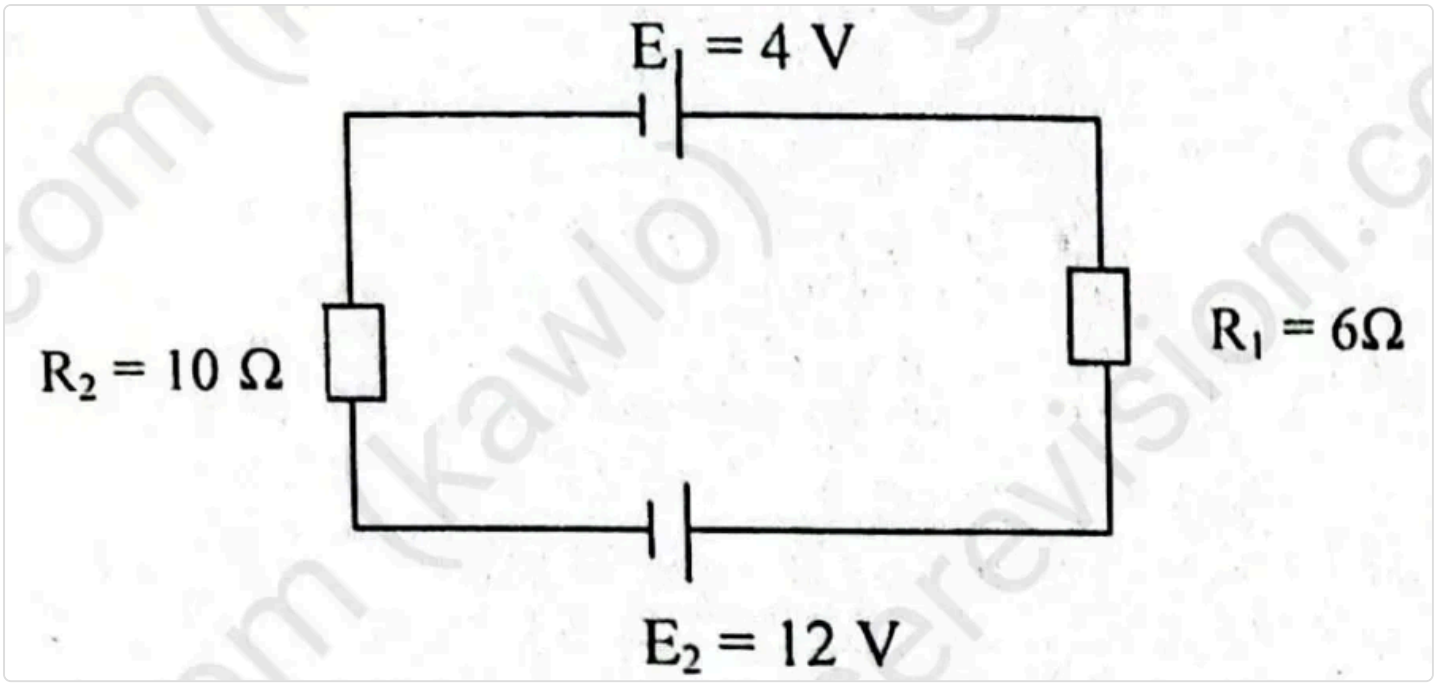


a) $1.5\ \mathrm{V}$
b) $3.75\ \mathrm{V}$
c) $2.5\ \mathrm{V}$
d) $4\ \mathrm{V}$

29. Which of the following quantities has units?

a) relative refractive index
b) relative velocity
c) relative permeability
d) relative permeability

30. Figure 6 shows an electric circuit which contains two resistors and two sources of emf. The internal resistances of the batteries are negligible. Which of the following is correct?

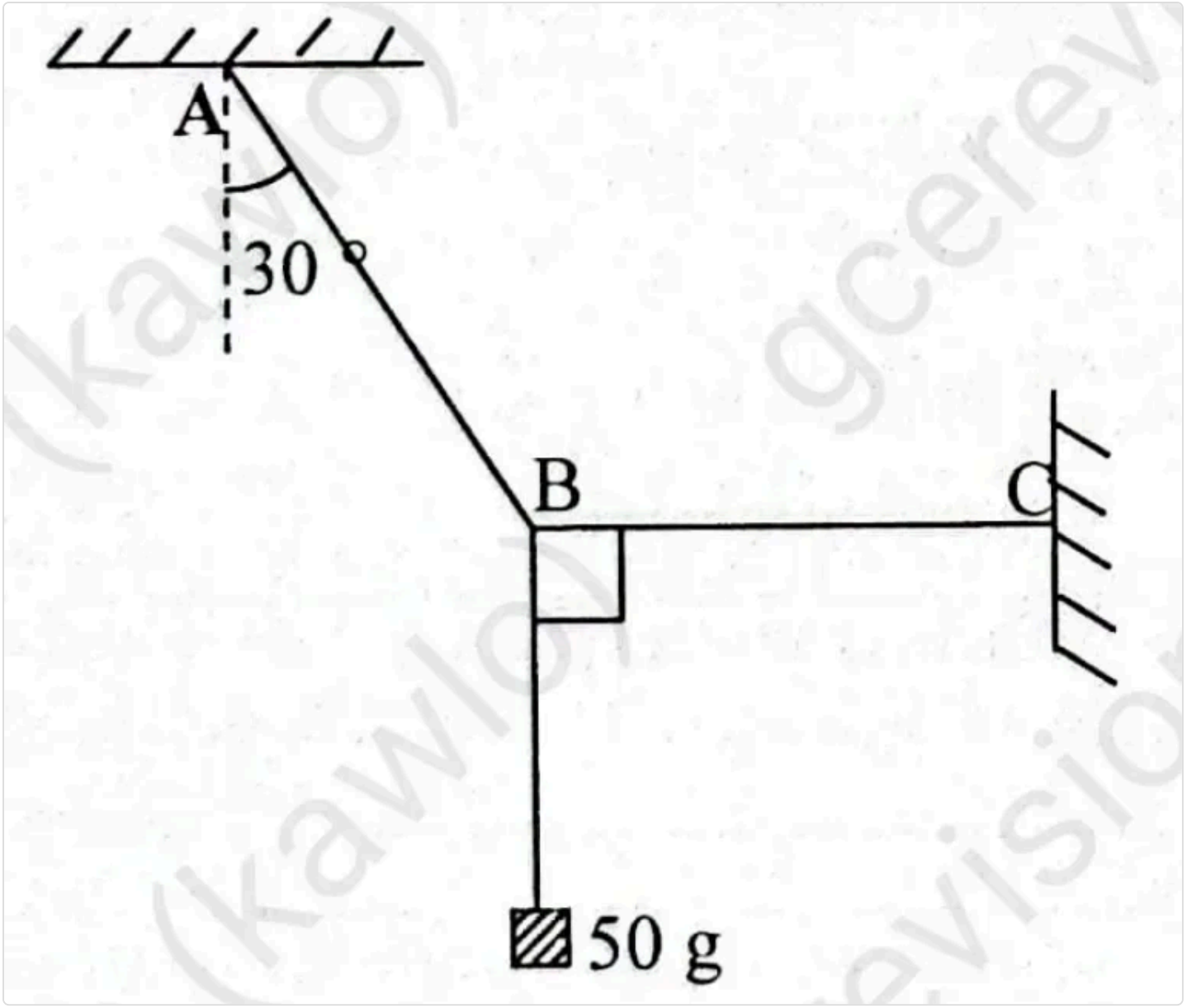


a) The power dissipated in R_1 is $1.5\ \mathrm{W}$
b) The pd across R_2 is $8\ \mathrm{V}$
c) The current in the circuit is $1.0\ \mathrm{A}$
d) The power dissipated in R_2 is $6\ \mathrm{W}$

31. A simple harmonic oscillator has a period of $0.01\ \mathrm{s}$ and amplitude of $0.2\ \mathrm{m}$. The maximum velocity in $\mathrm{m\,s^{-1}}$ of the oscillation is

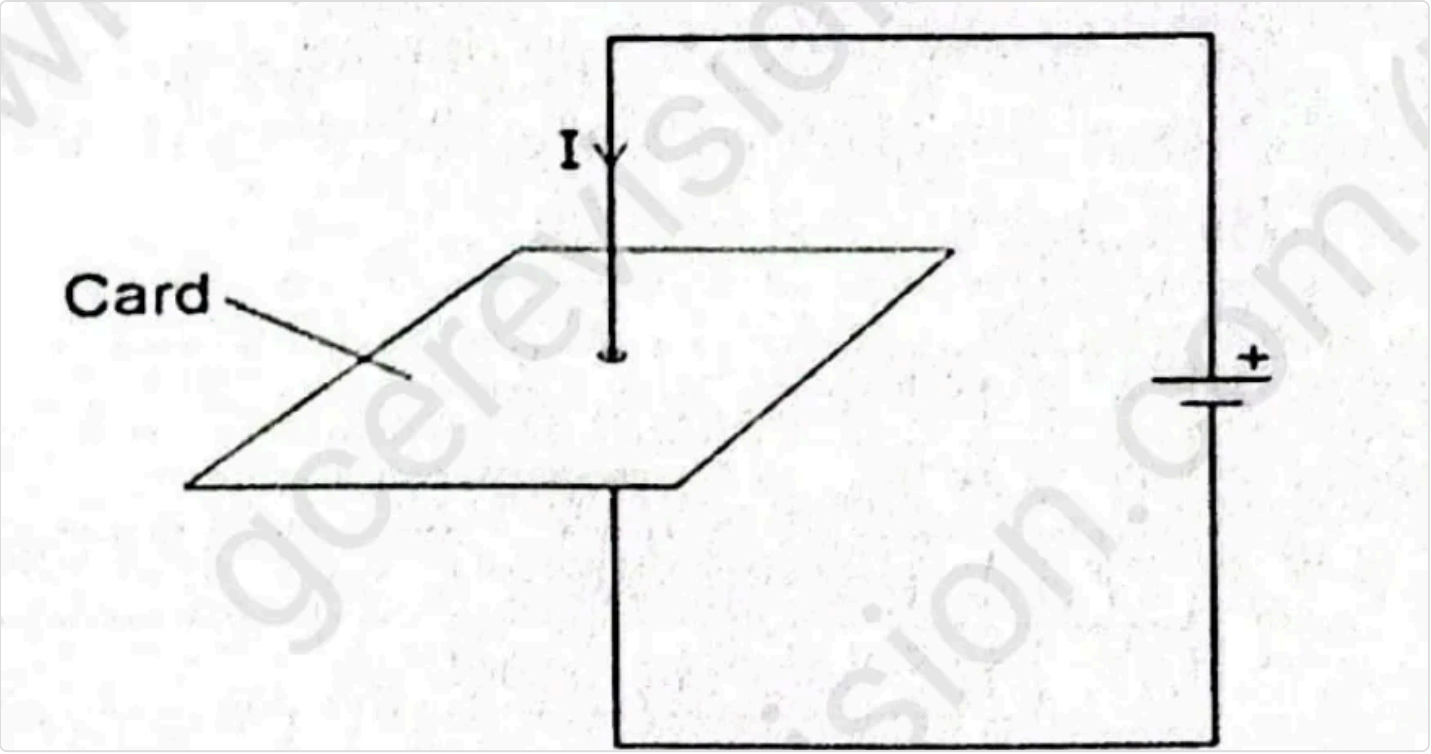
a) 126
b) 62.8
c) 25
d) 20

32. A mass of $50\ \mathrm{g}$ is attached to a spring which is fixed at supports, A and C as shown in figure 7. The tension in AB is



a) 0.98 N
b) 0.25 N
c) 0.42 N
d) 0.57 N

33. Figure 8 shows a straight wire passing through a piece of cardboard. The wire carries a current I. Which of the following is correct about the direction of the magnetic field formed?



a) It is clockwise as given by Lenz's law
b) It is anti-clockwise according to Fleming's left-hand rule
c) It is anti-clockwise according to Fleming's right-hand rule
d) It is clockwise according to Maxwell's corkscrew rule

34. The table below shows four energy levels of the hydrogen atom with their associated energies.

N	**Energy** / 10^{-19} J
1	-21.8
2	-5.3
3	-2.4
4	-1.3

The shortest wavelength line seen in the emission spectrum of hydrogen can result from a transition between which of these energy levels?

a) $n = 4$ to $n = 3$
b) $n = 3$ to $n = 1$
c) $n = 4$ to $n = 1$
d) $n = 2$ to $n = 1$

35. The pressure of a fixed mass of an ideal gas is doubled by heating at constant volume. Which of the following is true about the gas?

a) The density of the gas doubles.
b) The number of molecules per unit volume doubles.
c) The mean square speed of the molecules doubles.
d) The root mean square speed of the molecules doubles

36. An object is released from an aeroplane travelling horizontally at a height of $200\ \mathrm{m}$ above the ground. Which of the following statements is/are correct?

1. The initial velocity of the object relative to the plane is zero
2. The minimum time taken before the object hits the ground is $6.4\ \mathrm{s}$
3. The horizontal velocity of the object increases with time

Options as printed: A) 1 and 2 are correct B) 2 and 3 are correct C) 1 only is correct D) 3 only is correct

37. Which of the following observations of the photoelectric effect can be explained using the classical wave theory?

1. No time lapse between illumination and emission of electrons
2. Emission only occurs when the frequency of the light is above a certain threshold.
3. Number of emitted electrons per unit time is directly proportional to the light intensity

a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

38. In the S.I system of units, the unit of current, the ampere, is defined in terms of the force between two parallel conductors placed $1\ \mathrm{m}$ apart in a vacuum. Which of the following statements is/are true?

1. The magnitude of the force doubles if the current through each wire is doubled.
2. When a material of higher permeability is introduced into the space between the wires, the force increases slightly
3. The force acting per unit length of each wire is $2\pi \times 10^{-7}\ \mathrm{N\,m^{-1}}$

Options as printed: A) 1 and 2 are correct B) 2 and 3 are correct C) 1 only is correct D) 3 only is correct

39. Two waves of the same wavelength, λ, constant phase difference and same amplitude are emitted from X and Y, as shown in figure 9 above. At P there is a maximum; at Q a minimum and at R a maximum disturbance. Such an observation may be obtained if



1. $XP = YP$

2. $YQ - XQ = \lambda/2$
3. $YR - XP = \lambda/2$

a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

40. Which of the following types of damping would be most suitable for a moving coil galvanometer?

1. critical damping
2. heavy damping
3. light damping

a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

41. Which pair(s) of the following statements is/are true about progressive and stationary waves.

	For progressive waves	**For stationary waves**
1	There is no net transfer of energy	There is net transfer of energy
2	Amplitude is constant.	Adjacent points have different Amplitudes
3	There is gradual phase variation	Points between adjacent nodes are in phase

a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

42. Convex and concave lenses are image-forming devices. Which of these statements is/are correct about lenses?

1. For an object placed $20\ \text{cm}$ from a convex lens of focal length $4\ \text{cm}$, the image magnification is 0.25
2. For a convex lens, if the image is larger than the erect object, then the object distance is less than the focal length.
3. If an object is placed between a concave lens and the principal focus then the image formed is larger than the object.

Options as printed: A) 1 and 2 are correct B) 2 and 3 are correct C) 1 only is correct D) 3 only is correct

43. If a metal wire of cross-sectional area A, has n free electrons per unit volume each having a charge e and moving with a drift velocity v, then:

1. The total charge in the wire is ne
2. The current per unit area is nev
3. The rate of flow of charge is $nevA$

a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

44. Which of the following statements is/are true for a sinusoidal signal of peak voltage V_0 and Peak current I_0 flowing through a capacitor?

1. Average power dissipated through a resistive load is constant.
2. Current always leads voltage by $\pi/2$
3. The average power dissipated in a resistive load is $P_{av} = \dfrac{V_0 I_0}{\sqrt{2}}$

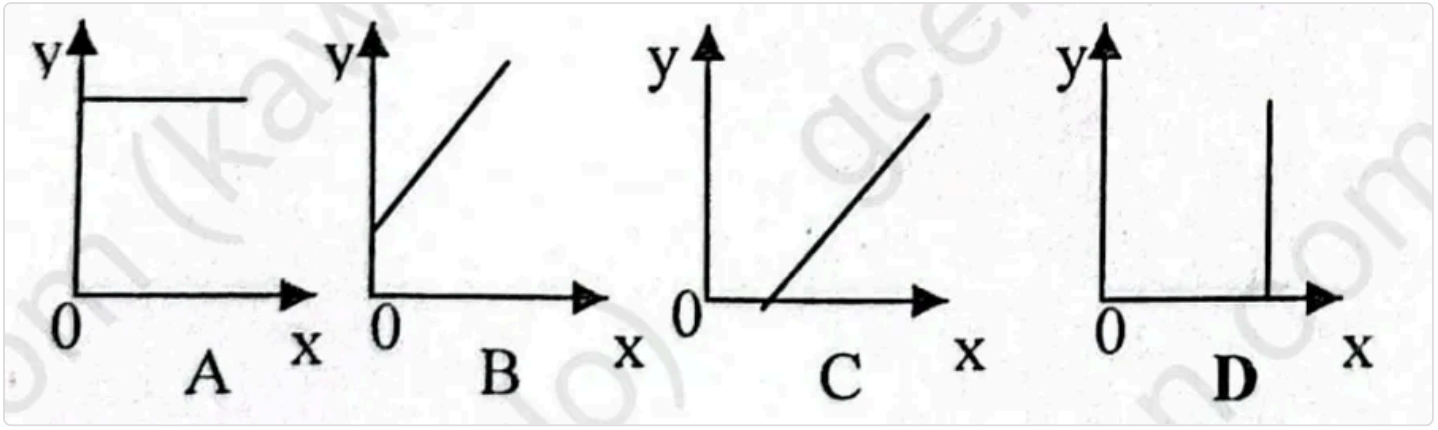
a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

45. If you dip your hand in alcohol and remove it, it feels cooler than the other fingers because;

1. Alcohol is a good conductor of heat
2. the alcohol is at a lower temperature than the atmospheric temperature
3. the alcohol molecules absorbs energy from the finger as they evaporate

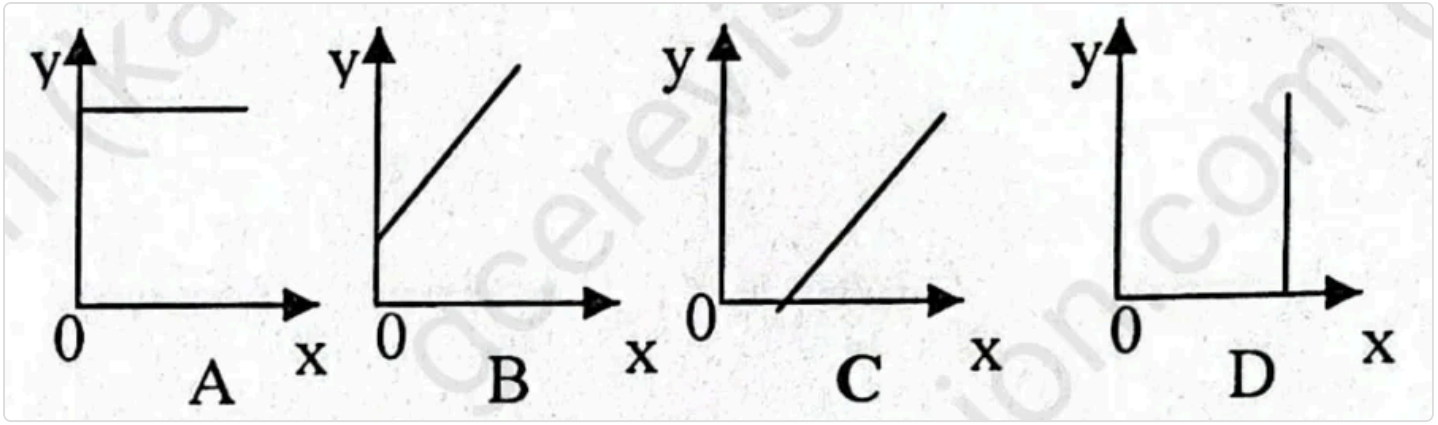
a) 1 and 2 are correct
b) 2 and 3 are correct
c) 1 only is correct
d) 3 only is correct

46. The graphs below are for y = average kinetic energy of molecules of a substance undergoing a change of state, and x = temperature of the substance.



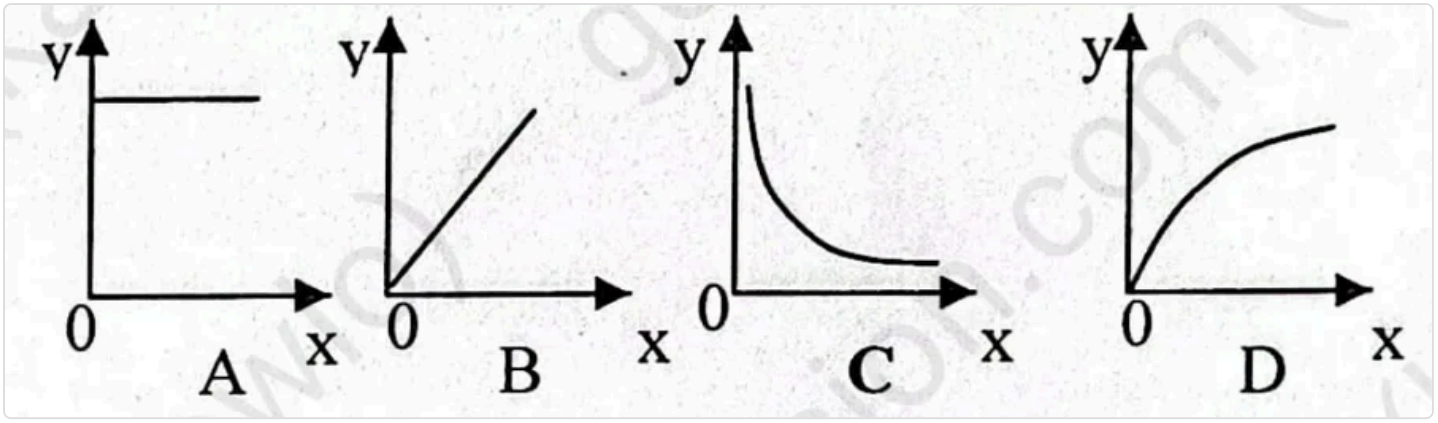
Options as printed: A) y constant (horizontal line) B) straight line rising from a positive y-intercept C) straight line rising from a positive x-intercept D) vertical line at a fixed value of x

47. The graphs below are for y = maximum kinetic energy of a photoelectron, and x = frequency of incident radiation.



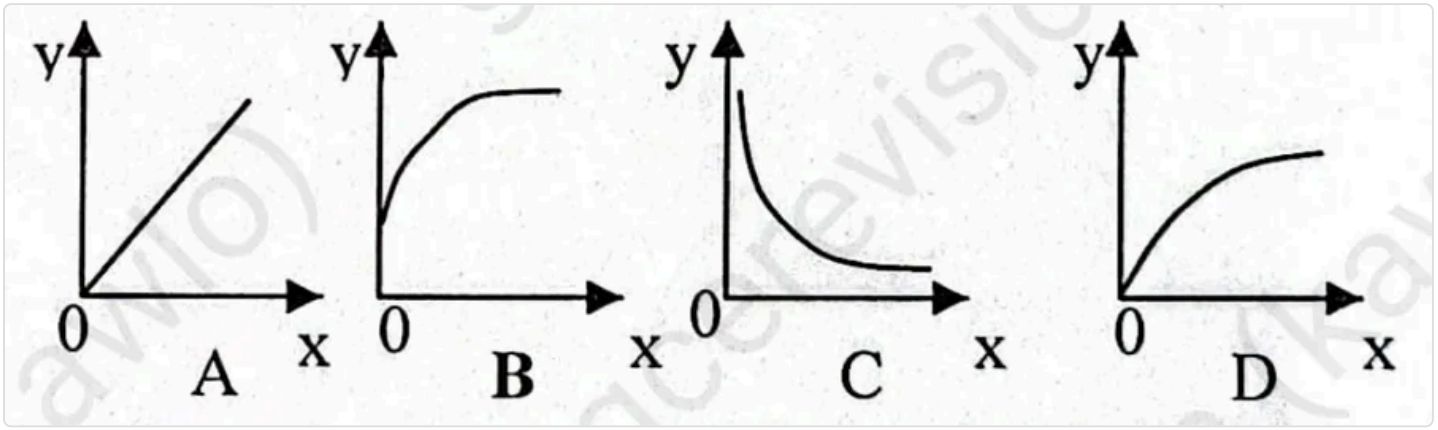
a) A
b) B
c) C
d) D

48. The graphs below are for y = potential energy of a planet orbiting the sun, and x = distance from the sun.



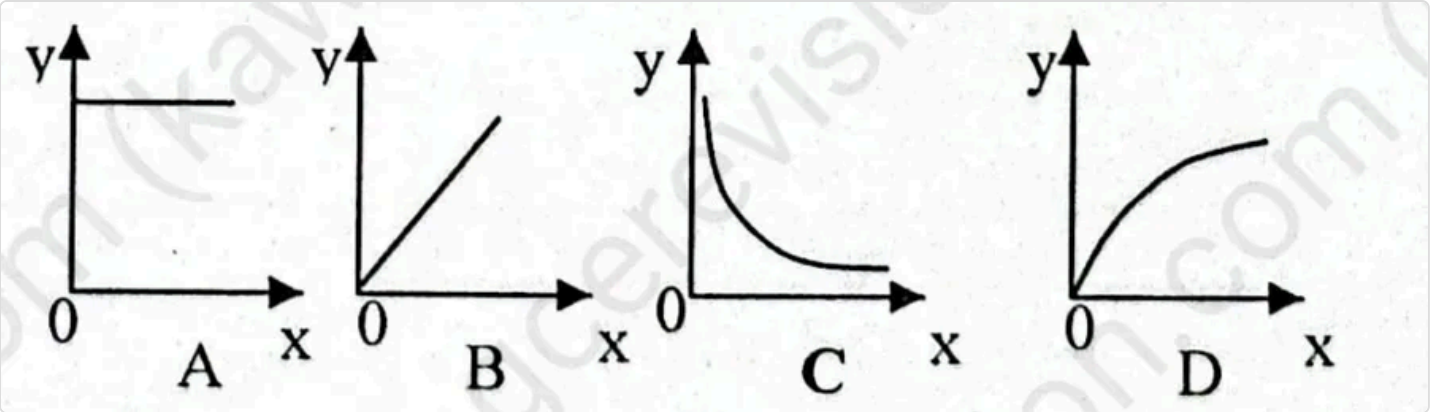
Options as printed: A) y constant (horizontal line) B) straight line through the origin C) decreasing curve (hyperbola) approaching a constant D) rising curve levelling off from the origin

49. The graphs below are for y = the magnitude of velocity of a box falling through air from an aeroplane in the sky, and x = the time of fall.



Options as printed: A) straight line through the origin B) rising curve levelling off, starting above zero C) decreasing curve (hyperbola) D) rising curve levelling off, starting from the origin

50. The graphs below are for y = the reactance of a capacitor, and x = frequency of a.c



a) A
b) B
c) C
d) D

Want to check your answers?

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